other means. To the extent firms automatically populate forms by using their own existing systems to create XML files, the EDGAR Filer Manual provides for submission of certain filings in a custom XML-based data language, which can be used to reach a similar result.[752] In addition, the structured data requirement enables EDGAR to perform technical validations (i.e., programmatic checks to ensure the documents are appropriately standardized, formatted, and complete) upon intake, thus improving the quality of the filed data by decreasing the incidence of errors (such as the omission of values from fields that should always be populated).[753] Furthermore, allowing firms to submit PDF documents would not achieve the benefits associated with the structured data requirements under the rule amendments to the same extent. The structured data requirements under the rule amendments will increase the accessibility and usability of the disclosures in the Structured Documents, for example by enabling more efficient retrieval, sorting, filtering, comparison, aggregation, and other analysis of the disclosures, thereby increasing transparency and insight into the operations, governance, management, financial condition, and other characteristics of the affected entities.

Structured Data Benefits

As an initial point of comparison, some research on XBRL requirements for public operating company financial statement disclosures has found that such requirements have

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752 See id. at 11 (suggesting, in the context of VDNs, that the Commission allow firms to submit a structured data file rather than filling in a web form); see supra section V.C.2 (discussing the option to use custom XML-based data language).

753 See, e.g., Semi-Annual Report to Congress Regarding Public and Internal Use of Machine-Readable Data for Corporate Disclosures (June 2024) at 4, https://www.sec.gov/files/fdta-report-6-2024.pdf (explaining that technical validation rules allow issuers to check for certain errors before the machine-readable data is submitted, which can streamline the compliance process by reducing Commission staff time that would otherwise be spent pinpointing and communicating the existence of technical errors to issuers, and by reducing issuer time that would otherwise be spent resolving such errors and resubmitting the machine-readable data file).

mitigated information asymmetry by reducing information processing costs, thereby facilitating access and analysis of the disclosures on a large-scale basis.[754] This reduction in information processing cost has been observed to facilitate the monitoring and analysis of firms by external parties.

These external parties include investors themselves, as well as other entities that process firm disclosures into conclusions that often influence investors and markets; such entities include financial analysts, data aggregators, academic researchers and financial media (collectively, "information intermediaries").[755] Institutional investors are more likely to access XBRL data directly, whereas retail investors are more likely to benefit from the use of XBRL data by information intermediaries.[756]

Regulators, including the Commission and the Federal Deposit Insurance Corporation ("FDIC"), have also been observed to leverage XBRL disclosure benefits in better fulfilling their

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754 See, e.g., Joung W. Kim, Jee-Hae Lim, & Won Gyun No, The Effect of First Wave Mandatory XBRL Reporting Across the Financial Information Environment, 26 J. INFO. SYS. 127, 127-53 (2012) (finding evidence that "mandatory XBRL disclosure decreases information risk and information asymmetry in both general and uncertain information environments").

755 See, e.g., N. Trentmann, "Companies Adjust Earnings for Covid-19 Costs, But Are They Still a One-Time Expense?", WALL ST. J. (2020) (citing an XBRL research software provider as a source for the analysis described in the article); Bloomberg Lists BSE XBRL Data, XBRL.ORG (Mar. 17, 2019), https://www.xbrl.org/news/bloomberg-lists-bse-xbrl-data/; R. Hoitash & U. Hoitash, Measuring Accounting Reporting Complexity with XBRL, 93 ACCOUNT. REV. 259 (Jan. 1, 2018).

756 See, e.g., A. Lawrence, et al., Investor Demand for Sell-Side Research, 92 ACCOUNT. REV. 123 (Mar. 1, 2017) (finding "the average retail investor appears to rely on analysts to interpret financial reporting information rather than read the actual filings"); but see S. Chi & D. Shanthikumar, Do Retail Investors Use SEC Filings? Evidence from EDGAR Search, (Nov. 8, 2018), available at https://ssrn.com/abstract=3281234 (retrieved from SSRN Elsevier database) (finding "retail investor trading, both buying and selling, is significantly related to EDGAR search for 10-K and 10-Q filings, more so than to Google search," especially for "the most easily readable 10-K and 10-Q filings"); see also N. Brown, et al., How do Disclosure Repetition and Interactivity Influence Investors' Judgments?, 60 J. ACCOUNT. RES. 1775 (Dec. 2022) ("Brown et al. iXBRL study") (indicating that disclosure interactivity, which is promoted by Inline XBRL, may improve investors' direct processing of financial information).

mandates.[757] The Commission staff uses XBRL data to efficiently analyze large quantities of information in support of risk assessment, rulemaking, and enforcement activities, including as part of its internally developed Financial Statement Query Viewer and Inline Viewer applications.[758] The regulatory use of XBRL is particularly relevant to affected documents that are subject to confidential treatment and thus only accessible by the Commission and its staff.[759]

The enhanced monitoring facilitated by XBRL requirements has been observed to influence the behavior of firms relevant to governance and compliance, including firms' disclosure and reporting choices. For example, one study found that firms increase quantitative footnote disclosures upon implementation of detailed tagging requirements.[760] Also, multiple

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757 With respect to Commission use of XBRL data, see infra note 758. With respect to FDIC use of XBRL data, see Meet Mark Montoya, Chief of Data Strategy, FDIC, XCENTIAL CO. (Sept. 29, 2021), https://xcential.com/blog/meet-mark-montoya-chief-data-officer-fdic/ (noting in an interview with the FDIC's Chief Data Officer that XBRL requirements for quarterly bank call reports have facilitated FDIC staff analysis of the regulated banks); see also Government Use of Data Standards – Conversation with the FDIC, XBRL US (Sept. 2, 2020), https://xbrl.us/news/regulator-video/ (noting in an interview with the FDIC's Chief Data Officer that, "…Prior (to XBRL) the data that the (FDIC) examiners used to examine the banks was probably about 2-3 months old which is old data . . . (with XBRL) the data can be pulled down in real time"); see also Lizhong Hao & Mark J. Kohlbeck, The Market Impact of Mandatory Interactive Data: Evidence from Bank Regulatory XBRL Filings, 10 J. EMERGING TECH. ACCT. 41 (Dec. 2013) (finding that banks experienced a "reduction in systematic risk in connection with filing their regulatory reports in XBRL").

758 See, e.g., Semi-Annual Report to Congress Regarding Public and Internal Use of Machine-Readable Data for Corporate Disclosures (June 2024), https://www.sec.gov/files/fdta-report-6-2024.pdf (describing Commission staff use of machine-readable data, including XBRL data, across various Divisions and Offices); T. Knutson, As XBRL in Financial Reporting Matures, Focus is on Accuracy, CFO DIVE (Feb. 25, 2020), available at https://www.cfodive.com/news/xbrl-financial-reporting-accuracy/572948/.

759 As stated above in sections X.A and X.B, whether any filed material is confidential is determined pursuant to applicable law, including but not limited to the Freedom of Information Act and Commission rules governing requests for confidential treatment.

760 See Elizabeth Blankespoor, The Impact of Information Processing Costs on Firm Disclosure Choice: Evidence from the XBRL Mandate, 57 J. ACCT. RES. 919 (2019) (finding "firms increase their quantitative footnote disclosures upon implementation of XBRL detailed tagging requirements designed to reduce information users' processing costs," and "both regulatory and non-regulatory market participants play a role in monitoring firm disclosures," suggesting "that the processing costs of market participants can be significant enough to impact firms' disclosure decisions"); see also Kim, Jeong-Bon, Kim, Joung W., &Lim, Jee-Hae, Does XBRL Adoption Constrain Earnings Management? Early Evidence from Mandated U.S. Filers, 36 CONTEMP. ACCT. RES. 2610 (2019) (indicating that XBRL adoption "constrains earnings management via discretionary accrual choices").

studies have shown that XBRL requirements have increased firms' investment efficiency by decreasing information processing cost and improved effective decision-making by managers, effects that appear to be heightened for Inline XBRL requirements.[761]

XBRL requirements have also been observed to impact the timeliness and effectiveness of firms' disclosure preparation and related processes. For example, one study found XBRL to have decreased audited report lags, especially among firms with strong internal control systems and no prior XBRL reporting experience.[762] Other studies have found XBRL requirements to have improved the timeliness of financial reports, with such improvements limited to larger firms only.[763] For instance, one public company executive stated that XBRL facilitates his firm's disclosure preparation procedures by enabling efficient review of disclosures made by peer companies.[764] Increasing the timeliness and effectiveness of the auditing and disclosure process

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761 See, e.g., Xin Cheng, et al., How Does Information Processing Efficiency Relate to Investment Efficiency? Evidence from XBRL Adoption, 35 J. INFO. SYS. 1 (2021) (finding firms "improve their investment efficiency after the adoption of XBRL," especially for firms that "have inferior external monitoring, . . . operate in more uncertain information environments, . . . and have less readable financial reporting"); see also Hyun Woong (Daniel) Chang, et al., The Effect of iXBRL Formatted Financial Statements on the Effectiveness of Managers' Decisions when Making Inter-Firm Comparisons, 35 J. INFO. SYS. 149 (2021) ("Chang et al. iXBRL study") (finding "iXBRL filings facilitate information search and information match by allowing users to view XBRL data in HTML filings," and "managers make more (less) effective decisions when presented with financial information formatted in iXBRL (XBRL)").

762 See K. Amin, J. Daniel Eshleman, & C. (Qian) Feng, The Effect of the SEC's XBRL Mandate on Audit Report Lags, 32 ACCT. HORIZ. 1 (Mar. 1, 2018) (finding "audit report lags decrease following the mandatory adoption of XBRL," with results "concentrated among filers with strong internal control systems and no prior XBRL reporting experience").

763 See, e.g., Hui Du & Kean Wu, XBRL Mandate and Timeliness of Financial Reporting: Do XBRL Filings Take Longer? 15 J. EMERG. TECH. ACCT. 57 (2018) (finding decreased reporting lags for XBRL annual and quarterly filings compared to non-XBRL filings from accelerated and large accelerated filers, but not for non-accelerated filers); see also J. Zhou, Does One Size Fit All? Evidence on XBRL Adoption and 10-K Filing Lag. 60 ACCT FIN. 3183 (Sept. 2020) (noting that 10-K filing lag decreased for all filers in the XBRL reporting period except smaller reporting companies, for which 10-K filing lag increased). However, these studies were based on XBRL filings that were made before the adoption of Inline XBRL requirements, which may facilitate the filing preparation process by including the machine-readable and human-readable data in the same disclosure document.

764 See Olivia Berkman, XBRL: What are the Benefits, FEI DAILY (Aug. 29, 2019), https://www.financialexecutives.org/FEI-Daily/August-2019/XBRL-What-are-the-Benefits.aspx (noting in

would improve the speed (and, with respect to enhanced auditing processes, confidence) with which users of the affected entities' disclosures (such as investors, analysts, and regulators) could assess and ultimately draw conclusions from, and act upon, the disclosed information.[765]

One commenter stated that the empirical evidence the Commission provided to justify the use of XBRL and XML was limited to a single study that does not analyze Inline XBRL. However, the discussion above cites multiple studies, not a single study, providing evidence of benefit from XBRL.[766] Further, while most of the cited studies discuss XBRL rather than Inline XBRL (presumably because the phase-in from XBRL requirements to Inline XBRL requirements for Commission filings was completed relatively recently, in 2021), two of the cited studies discuss Inline XBRL specifically.[767]

Applicability and Variability of Structured Data Benefits

The structured data benefits discussed above, while largely specific to public operating company financial statement disclosures, generally indicate that the structured data requirements under the rule amendments will facilitate the use and analysis of the information disclosed on the affected documents. One commenter agreed with this statement, stating that data processing is significantly faster with XBRL than with unstructured data types.[768] Several of the affected documents that are required to be structured in Inline XBRL—namely, Form X-17A-5 Part III,

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an interview with a public company's chief financial officer that the company is able to "search through XBRL filings to find similar companies within [its] industry that have had to present certain similar [disclosures] in the past," which has helped the company "craft[] [its] disclosures to make sure that [the company is] complying with the spirit of GAAP and providing the information that [the company is] supposed to be providing").

765 See Proposing Release at 23946 (discussing the time lag between the date of receipt of a paper filing of a broker-dealer's annual reports until it is scanned and the public portion published on EDGAR, and the confidential portion available to Commission staff).

766 See supra notes 754 to 756 and 760 to 763.

767 See Brown et al. iXBRL study; Chang et al. iXBRL study.

768 See XBRL Letter at 11.

Form 17-H, Form 1, and Form CA-1—include financial statements that were not previously provided in a structured data language, but will be provided in a structured data language (specifically, Inline XBRL) under the rule amendments. The probability that, and extent to which, the observed effects can be extrapolated are thus likely greater for those affected documents than for the remaining affected documents, which do not contain financial statements.

In addition, unlike the public company financial statement information evaluated in the literature referenced above, several of the affected documents are submitted confidentially or are otherwise non-public, either in whole or in part. This includes Form 17-H, Form X-17A-19, Form X-17A-5 Part III (in part), Form CA-1 (in part), and the CCO reports.[769] The expected benefits of structuring non-public information will accrue to investors and markets indirectly, by enhancing the Commission's regulatory capabilities.[770] By contrast, the expected benefits of structuring public information will accrue directly to public users of the data (which could include investors and the previously discussed information intermediaries), as well as indirectly to investors and markets through the enhancement of the Commission's regulatory capabilities (and, where relevant, those of other regulators).

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769 Additionally, the Commission does not automatically make public the information provided to it pursuant to Rule 15fi-3(c); however, the Commission may make the information available upon appropriate request (including requests made pursuant to the Freedom of Information Act) or otherwise as permitted under applicable law, subject to SBS Entities making appropriate requests for confidential treatment. See supra section IX.F. Whether any material is confidential is determined pursuant to applicable law, including but not limited to the Freedom of Information Act and Commission rules governing requests for confidential treatment. If the Commission makes the information provided pursuant to Rule 15fi-3(c) available, the information made available may not be in structured format.

770 See supra sections II.D, IV.A, IV.B, V.A, and V.D. An example of a structured non-public disclosure form is Form PF, which registered investment advisers file with the Commission to disclose information regarding private funds under their management. See 17 CFR 275.204(b); Division of Investment Management: Electronic Filing of Form PF for Investment Advisers on PFRD, available at https://www.sec.gov/divisions/investment/pfrd.shtml.

The benefits of structuring will also vary based on the number of entities in a particular population of reporting entities. As stated, one benefit of structured disclosure is the ability to run large-scale comparisons across reporting entities and across reporting periods. For those affected documents that have a small population of reporting entities, this benefit will be limited largely (or, in the case of Form 15A, wholly) to the latter. For those affected documents that have a large population of reporting entities (such as Form X-17A-5 Part III, which is filed by over 3,000 entities), the benefits of efficient cross-entity comparisons will be much more relevant.[771] Similarly, the benefits of efficient access, retrieval, sorting, and filtering structured disclosures will be heightened for those affected documents generated in high volume (such as Form 19b-4(e) and Form X-17A-19) compared to those affected documents that the Commission receives in low volume (such as Form CA-1).[772]

Finally, the benefits of structuring data will vary based on the type of disclosures included in each affected document. Structured numerical disclosures lend themselves to mathematical functionality, such as the calculation of leverage or other ratios to assess potential exposure to insolvency or other risk. Structured textual disclosures lend themselves to period-over-period redline comparisons, targeted keyword searching, and more sophisticated sentiment analysis. The CCO report consists primarily of textual descriptions, so the latter benefit will be relevant for that document.[773] Other affected documents feature both numeric and textual disclosures, so both benefits will be relevant.

Types of Content and Associated Benefits in Structured Documents

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771 See supra section IX.C.9.

772 See supra sections IX.D.5, IX.D.6, and IX.D.11.

773 See 17 CFR 240.15fk-1(c)(2)(i). SBS Entities may, in providing the narrative descriptions required by Rule 15Fk-1(c)(2)(i), include numeric values nested within such narrative descriptions.

Structured Document	Numeric Information (mathematical functionality applicable)	Textual Information (redline comparisons, targeted searches, sentiment analyses applicable)
Form X-17A-5 Part III	Yes	Yes
Form 17-H	Yes	Yes
Form CA-1	Yes	Yes
Form 1	Yes	Yes
Form 1-N (execution page only)	No	Yes
Form 15A (execution page only)	No	No
Rule 19b-4(e) Information	In some cases	No
Form X-17A-19	No	Yes
VDN	Yes	Yes
CCO report	In some cases	Yes

One commenter disagreed with the Commission's characterization of benefits for structuring the CCO report (as well as other unstructured, narrative-based reports such as the compliance and exemption sections of the annual audited report on Form X-17A-5 Part III).[774] According to this commenter, adding an Inline XBRL requirement for narrative reports will not facilitate analysis or comparison, because those reports do not contain standardized, easily comparable elements.[775] However, all narrative reports must include disclosure responsive to applicable disclosure requirements set forth in the Commission's rules and regulations (e.g., the disclosure requirements set forth in the subparagraphs of Rule 15fk-1(c)(2)(i) under the Exchange Act). While there may be variation on the particulars on how different filers or submitters respond to those requirements, the Inline XBRL requirement will facilitate analysis

774 See SIFMA 5/22/2023 Letter at 7.

775 See id.

by enabling efficient assessment of such variations, and by enabling efficient comparisons of a single filer or submitter's narrative disclosure over various time periods, allowing the data user to determine how that filer's or submitter's disclosure has evolved over time.

The same commenter also questioned why the Commission cited sentiment analysis as a benefit of Inline XBRL requirements.[776] According to the commenter, sentiment analysis is typically used for marketing purposes, and thus it was not clear why such analysis would be necessary or beneficial for narrative reports. However, sentiment analysis is often used for purposes beyond marketing, including by third parties to assess regulatory disclosures, such as disclosures in Commission filings in order to assess the usefulness of disclosures to end users in the market.[777] Thus, the sentiment analysis benefit is applicable to the narrative reports that are structured in Inline XBRL under the rule amendments.

For Rule 19b-4(e), numeric disclosures are required only when the disclosure of position limits for new derivative securities products is applicable.[778] For VDNs, SBS Entities must notify the Commission of any valuation disputes in excess of $20,000,000 if not resolved within three or five business days, depending on the counterparty.[779] SBS Entities are provided flexibility to submit the required information.[780] For CCO reports, while Rule 15fk-1(c) does not expressly call for numeric values, an SBS Entity could include numeric values nested within textual responses, such as by including dollar amounts within the description of financial,

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776 See id.

777 See, e.g. H. Kim, E. Lee, & D. Yoo, Do SEC Filings Indicate Any Trends? Evidence from the Sentiment Distribution of Forms 10-K and 10-Q with FinBERT, 57 DATA TECH. & APPLICATIONS 293 (Apr. 25, 2023); see generally, Yong Chen et al., Sentiment Trading and Hedge Fund Returns, 76 J. Fin. 2001 (Apr. 8, 2011).

778 See Item 9 of Form 19b-4(e).

779 See 17 CFR 240.15fi-3(c)(1). See also 17 CFR 240.15fi-3(c)(2) regarding required amendments.

780 See supra section V.C.

managerial, operational, and staffing resources set aside for compliance with the Exchange Act.[781]

For Form 15A, its execution page (i.e., the portion of new Form 15A that would precede section I) will include a series of structured checkboxes to indicate the basis for filing the Form, and the reporting period to which the Form applies. Structured checkboxes and pick lists are more relevant to the filtering and sorting benefits enabled by structured data requirements. For example, structuring the checkboxes on the Form 15A execution page will enable a data user to retrieve only those Form 15As that are current supplements to registration reported pursuant to Rule 15aa-2(b) of the Exchange Act, and further filter those results to only those Form 15A filings that include a change to Exhibit C (list of members).

The Commission is requiring a specific structured data language for each Structured Document, rather than leaving the structured data language requirement open-ended (i.e., requiring only that the Structured Document be provided in a structured, machine-readable data language). Specifying a single structured data language that a filer or submitter must use for each Structured Document will benefit users of the disclosed information, including investors, market participants, other filers or submitters, information intermediaries, and the Commission, because it will help ensure the disclosures are provided in a uniform structured data language that is most suitable for the document in question, and will prevent a potential coordination failure that could occur if different respondents chose to provide inputs in different data languages.

One commenter disagreed with requiring a specific structured data language for each Structured Document, stating that the Commission should instead adopt a principles-based

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[781] See 17 CFR 240.15fk-1(c)(2)(i)(E).